Myocardial Injury After Noncardiac Surgery and its Association With Short-Term Mortality

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Background—To identify patients at risk for postoperative myocardial injury and death, measuring cardiac troponin routinely after noncardiac surgery has been suggested. Such monitoring was implemented in our hospital. The aim of this study was to determine the predictive value of postoperative myocardial injury, as measured by troponin elevation, on 30-day mortality after noncardiac surgery.

Methods and Results—This observational, single-center cohort study included 2232 consecutive intermediate- to high-risk noncardiac surgery patients aged ≥60 years who underwent surgery in 2011. Troponin was measured on the first 3 postoperative days. Log binomial regression analysis was used to estimate the association between postoperative myocardial injury (troponin I level >0.06 μg/L) and all-cause 30-day mortality. Myocardial injury was found in 315 of 1627 patients in whom troponin I was measured (19%). All-cause death occurred in 56 patients (3%). The relative risk of a minor increase in troponin (0.07–0.59 μg/L) was 2.4 (95% confidence interval, 1.3–4.2; $P<0.01$), and the relative risk of a 10- to 100-fold increase in troponin (≥0.60 μg/L) was 4.2 (95% confidence interval, 2.1–8.6; $P<0.01$). A myocardial infarction according to the universal definition was diagnosed in 10 patients (0.6%), of whom 1 (0.06%) had ST-segment elevation myocardial infarction.

Conclusions—Postoperative myocardial injury is an independent predictor of 30-day mortality after noncardiac surgery. Implementation of postoperative troponin monitoring as standard of care is feasible and may be helpful in improving the prognosis of patients undergoing noncardiac surgery. **(*Circulation*. 2013;127:2264-2271.)**

Key Words: ischemia ■ mortality ■ myocardial infarction ■ surgery ■ surveillance

Perioperative adverse cardiovascular events are the leading cause of morbidity and mortality after noncardiac surgery.[1] Despite efforts to prevent the occurrence of such events,[2,3] the incidence of postoperative myocardial infarction (POMI) after noncardiac surgery is still high and varies depending on the definition used.[4,5] With the use of the latest universal definition of myocardial infarction in a large cohort of patients undergoing noncardiac surgery, the incidence of POMI was as high as 5%.[4–6] Because symptoms such as chest pain are easily masked by adequate postoperative pain management including opioids, the clinical course of POMI is mainly silent.[6] As a result, POMI is recognized late or not recognized at all but remains strongly associated with mortality.[1,6–9] The mortality rate in patients with asymptomatic POMI is similar to that of patients who experience ischemic symptoms.[6] Moreover, even patients with minor elevations of troponin after surgery, without any symptoms of ischemia or signs of myocardial infarction, seem to have a comparably high risk of cardiovascular complications and death.[6,10–13]

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Although the pathophysiology of such isolated postoperative troponin elevation is not yet fully understood,[1,14] a meta-analysis of studies in predominantly high-risk surgery patients showed that isolated troponin elevation was a strong independent predictor of mortality within the first year after surgery.[10] Of note, troponin elevation was associated with a 6-fold increase in mortality.[10] In addition, the recently published Vascular Events in Noncardiac Surgery Patients Cohort Evaluation (VISION) study including >15 000 surgical patients showed a strong association between peak troponin levels after surgery and 30-day mortality.[11] Time to death after the peak troponin values was reported to vary between 1 and 2 weeks, potentially allowing for interventions to alter a patient's risk of death.[11] The accumulating evidence from VISION and other studies suggests that implementation of postoperative troponin monitoring as standard of care in patients undergoing noncardiac surgery may be useful

Received July 12, 2012; accepted April 12, 2013

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Circulation **is available at http://circ.ahajournals.org** **DOI: 10.1161/CIRCULATIONAHA.113.002128**

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for better risk stratification and management of patients.[6,10–13]

Therefore, we implemented routine troponin measurement monitoring on the first 3 days after surgery as part of our standard postoperative care in patients aged ≥60 years undergoing all types of intermediate- to high-risk noncardiac surgery. We hypothesized that the results of this routine monitoring would improve risk stratification of patients at risk for early postoperative death. Therefore, the aim of the present study was to determine the predictive value of postoperative troponin elevation as a marker of myocardial injury in relation to 30-day mortality after noncardiac surgery.

Methods

Patients

This observational cohort study included consecutive patients undergoing noncardiac surgery between January 3, 2011, and December 15, 2011, at the University Medical Center Utrecht, Utrecht, Netherlands, a 1000-bed tertiary referral hospital. Patients were eligible if they were aged ≥60 years, were undergoing intermediate- to high-risk noncardiac surgery under general or spinal anesthesia, and had an expected postoperative length of hospital stay of at least 24 hours. For patients who underwent surgery more than once during the study period, the first surgery was included in the analyses for all patients. A reoperation was included as a novel case if this surgery took place during another hospital admission and at least 30 days after the first surgery. Patients were excluded if they were lost to follow-up within 30 days after surgery.

The local medical ethics committee approved the study protocol. The need for informed consent was waived because only routinely collected patient data were used and data were anonymized before analysis (UMC Utrecht Medical Research Ethics Committee 11-120/C).

Data Collection

All preoperative and postoperative data were obtained from electronic medical and administrative records. Data collected in all patients included patient characteristics, preoperative physical status, comorbidities, and death within 30 days. In those patients with postoperative troponin I measurements, troponin I levels were collected. Additionally, data on postoperative symptoms, ECG changes, incidence of POMI, and treatment initiated by the consultant cardiologist were collected in those patients who had a cardiology consultation. The unique hospital patient identifier was used to merge databases. For data on 30-day mortality, the municipal personal records database was consulted.

Troponin Measurements

According to the hospital protocol, cardiac troponin I measurements were ordered for the first 3 days after surgery by the attending anesthesiologist immediately after the case was completed in the operating department. In case of a troponin I elevation above the clinical cutoff level during the first 3 days after surgery, the ward physician was notified. Because the optimal treatment of patients with postoperative troponin elevation is not yet known and must be interpreted in a manner consistent with a patient's clinical course, it was left to the discretion of the treating physician whether further diagnostic procedures, including an ECG or a consultation with the cardiologist, were indicated in individual patients. In this case, troponin elevation was considered simply a marker for myocardial injury, warranting additional attention.

In this study, myocardial injury was defined as a troponin I >0.06 μg/L, which was the lowest value measurable with a 10% coefficient of variation above the 99th percentile of 0.04 μg/L of the assay used.[15] Troponin I was analyzed with the use of the third-generation enhanced AccuTnI assay (Beckman Coulter, Brea, CA). For each patient, the highest value of all routine troponin I measurements was used in the analysis.

Outcomes

The primary outcome was defined as all-cause mortality within 30 days after surgery. Secondary outcomes included the incidence of POMI and length of hospital stay. Myocardial infarction was defined according to the universal definition.[15]

Statistical Analysis

The analysis was performed with the use of SPSS (release 20.0 for Windows). In advance of data analysis, we were aware that troponin I results would not be available in all patients because adherence to a new protocol in daily clinical practice takes time. Therefore, we compared baseline characteristics and incidence of death in patients in whom troponin I was measured with those in whom it was not to determine whether the missing troponin I measurements could be considered missing at random. To check whether excluding the missing values from the analysis influenced the association between postoperative myocardial injury and death, we repeated the multivariable regression analysis after multiple imputations of the missing troponin I values. Five data sets were imputed by the method of fully conditional specification. The baseline characteristics were used as predictors of the troponin I values. In the primary analysis, we first examined the distribution of the first troponin I elevation over the postoperative days. Second, the primary outcome and the incidence of POMI were compared for patients with and without myocardial injury with the χ^2 test, and relative risks with 95% confidence intervals (CIs) were calculated. In this case, troponin I was included as both a continuous and a categorical variable. In the latter, troponin I was categorized into 4 groups on a logarithmic scale, as follows: troponin I ≤0.06, 0.07 to 0.59, 0.60 to 5.99, and ≥6.00 μg/L. The median time to death was calculated, and the median length of stay in the hospital of both groups was compared with the Mann-Whitney test. Kaplan-Meier survival analysis was used to determine the survival of patients in each category of troponin I elevation. Subsequently, a log binomial multivariable regression model was developed to adjust the association between myocardial injury and death for preoperative variables known to predict postoperative cardiovascular events.[16] We used log binomial regression analysis to facilitate presenting effect measures as relative risks. In this case, we used model-based covariance estimates. The model included troponin I and age, sex, emergency surgery, and renal failure (preoperative glomerular filtration rate <50 mL/h).[16] Emergency surgery was defined as surgery required within 72 hours. Because the number of patients with a troponin I level >6.00 μg/L was considered too small to obtain relevant results, in this analysis the 2 upper troponin I categories were merged into 1 (ie, with troponin I ≥0.60 μg/L). We compared the predictive value of a model with only preoperative variables with a model in which troponin I was added to these variables using the area under the receiver operating characteristic curve and integrated discrimination improvement analysis. We preferred the latter method over using the net reclassification improvement because it does not require (arbitrary) classification of the mortality risk.[17]

Results

In total, 2232 patients were eligible for inclusion. Sixteen patients (0.7%) were lost to follow-up and excluded from the analysis, as follows: Three patients were transferred to another hospital on the day of surgery, 6 patients died on the day of surgery (ie, troponin I could not be measured in these patients), and in 7 patients mortality data could not be obtained because the patients were not known by the municipal personal records database. Baseline characteristics of the 2216 remaining patients are given in Table 1. Troponin I was measured in 1627 patients (73.4%). It was measured consecutively on all 3 postoperative days in 907 patients (55.7%) and was elevated at least once in 315 patients (19.4%). Within the group of patients with myocardial injury, the first elevated troponin I level was found on the first postoperative day in 162 patients

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(51.4%; Figure 1). Stratification of the patients into decades of age showed that the incidence of postoperative myocardial injury increased with age (Figure 2).

Postoperative myocardial injury was associated with an increased risk of death. Twenty-seven of the 315 patients (8.6%; 95% CI, 6.0–12.2%) with myocardial injury died within 30 days compared with 29 of the 1312 patients (2.2%; 95% CI, 1.5–3.2%) with normal troponin I levels ($P<0.01$).

Table 1. Baseline Characteristics

	Troponin Measured (n=1627)		Troponin Not Measured (n=589)		*P* Value
Male	844	(51.9)	303	(51.4)	0.86
Age, mean (SD), y	71.0	(7.7)	70.2	(7.4)	0.02
RCRI factors					
High-risk surgery	501	(30.8)	120	(20.4)	<0.01
History of ischemic heart disease	264	(16.2)	80	(13.6)	0.13
History of heart failure	35	(2.2)	16	(2.7)	0.43
History of cerebrovascular disease	242	(14.9)	88	(14.9)	0.97
Renal failure	166	(10.2)	68	(11.5)	0.36
Preoperative insulin use	92	(5.7)	35	(5.9)	0.80
Smoking*	286	(18.2)	97	(17.4)	0.68
Hypertension	829	(51.0)	291	(49.4)	0.52
Diabetes mellitus	288	(17.7)	106	(18.0)	0.87
COPD	167	(10.3)	48	(8.1)	0.14
Peripheral vascular disease	180	(11.1)	46	(7.8)	0.03
ASA class†					
1	197	(12.1)	89	(15.1)	0.06
2	1043	(64.1)	357	(60.6)	0.13
3	366	(22.5)	141	(23.9)	0.48
4	21	(1.3)	2	(0.3)	0.05
General anesthesia	1514	(93.1)	537	(91.2)	0.14
Emergency surgery	306	(18.8)	157	(26.7)	<0.01
Reoperation within 30 d	163	(10.0)	49	(8.3)	0.23
Surgical specialty					
General	369	(22.7)	113	(19.2)	0.08
Neurological	355	(21.8)	123	(20.9)	0.64
Vascular	255	(15.7)	76	(12.9)	0.11
ENT and dental	206	(12.7)	76	(12.9)	0.88
Orthopedic	193	(11.9)	96	(16.3)	<0.01
Gynecological	128	(7.9)	46	(7.8)	0.97
Urological	84	(5.2)	36	(6.1)	0.38
Plastic	37	(2.3)	23	(3.9)	0.04

Values are number of patients (%), unless indicated otherwise.

ASA indicates American Society of Anesthesiologists; COPD, chronic obstructive pulmonary disease; ENT, ear, nose, and throat; and RCRI, Revised Cardiac Risk Index.

*n=1568 and n=556, respectively, because of some missing values for smoking status.

†Physical status classification by the American Society of Anesthesiologists.

The median time to death after troponin I elevation was 12 days (interquartile range, 13). Patient characteristics of the 3 categories of troponin elevation are given in Table 2. The unadjusted relative risk of death was 3.0 in case of a minor increase in troponin I (0.07–0.59 μg/L) and 7.9 in case of a ≥10-fold increase in troponin I (≥0.60 μg/L; Table 3). The survival in each of these categories is shown in Figure 3. The analysis including troponin I on a continuous scale showed an increase in relative risk for each 0.1-μg/L increase in troponin I level (relative risk, 1.003; 95% CI, 1.001–1.004).

After adjustment of the association between myocardial injury and death for variables known to predict postoperative cardiovascular events, only emergency surgery and troponin I remained significantly related to death within 30 days. The relative risk of death was 2.4 (95% CI, 1.3–4.2; $P<0.01$) in case of a minor increase in troponin I (0.07–0.59 μg/L) and 4.2 (95% CI, 2.1–8.6; $P<0.01$) in case of a ≥10-fold increase in troponin I (≥0.60 μg/L; Table 3). The area under the receiver operating characteristic curve of the model with only preoperative predictors was 0.75 (95% CI, 0.68–0.83) compared with 0.78 (95% CI, 0.71–0.85) for the model in which troponin I was added. The integrated discrimination improvement index of the model including troponin I was 0.02 ($P<0.001$).

The median length of stay in the hospital was 10 days (interquartile range, 12) in patients with myocardial injury compared with 5 days (interquartile range, 6) in patients without myocardial injury ($P<0.01$).

Ten of the 315 patients (3.2%) with troponin elevation had typical chest pain, and ECG changes suggestive of new ischemia were found in 30 patients (9.5%). One (0.3%) of these ECGs showed ST-segment elevation (≥1 mm), 14 (4.4%) showed ST-segment depression (≥1 mm), and 15 (4.8%) showed minimal ST-segment depression (<1 mm) or repolarization changes.

A cardiologist was consulted in 110 of the 315 patients (34.9%). In most of these 110 patients, the clinical course was awaited without any intervention (n=67; 60.9%), and in 43 patients (39.1%), cardiovascular medication was changed or added (statins and aspirin). Eight patients (7.3%) were transferred to a coronary care unit for continuous ECG monitoring. Coronary angiography was performed in 7 patients (6.4%), and significant coronary stenoses were found in 6 (5.4%) of these patients. A percutaneous coronary intervention was performed in 4 patients (3.6%), 1 patient (0.9%) underwent coronary artery bypass surgery, and in 1 other patient (0.9%) a coronary intervention was considered not to be beneficial because of the patient's poor clinical condition.

POMI according to the criteria of the universal definition was diagnosed in 10 of the 1627 patients (0.6%). This concerned a non– ST-segment elevation myocardial infarction in 9 patients (0.6%) and an ST-segment elevation myocardial infarction in 1 patient (0.06%). Because the ST-segment elevation myocardial infarction was considered to have been sustained for >6 hours, coronary angiography was not performed immediately. This patient was initially treated with medication, and percutaneous coronary intervention for a nonsignificant coronary stenosis was performed in an outpatient setting.

To determine whether the missing troponin I measurements could be considered missing at random, patients in whom

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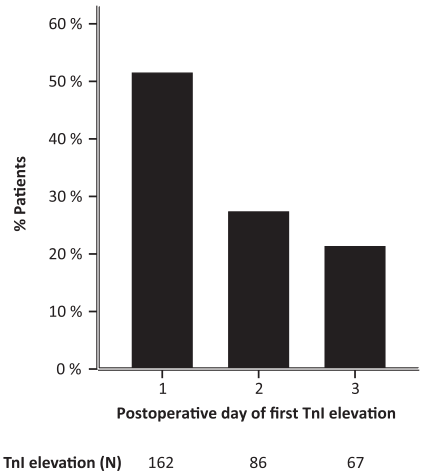


Figure 1. Distribution of the first troponin elevations over the first 3 postoperative days. TnI indicates troponin I.

troponin I was measured were compared with those in whom it was not. Death occurred in 56 of the 1627 patients (3.4%; 95% CI, 2.7–4.4) in whom troponin was measured compared with 20 of the 589 patients (3.4%; 95% CI, 2.2–5.2) in whom troponin was not measured (P=0.96). After imputation of the missing troponin I values, multivariable regression analysis showed a comparable association between postoperative myocardial injury and death (Table 4).

Discussion

This cohort study estimated the incidence of myocardial injury after noncardiac surgery and its predictive value on the risk of death within 30 days with the use of data from routine postoperative monitoring of troponin. Troponin I was elevated in 19% of the patients, and troponin I elevation was an independent predictor of 30-day mortality. This association with the risk of death was dependent on the degree of the troponin I elevation (relative risk between 2.4 and 4.2) but independent

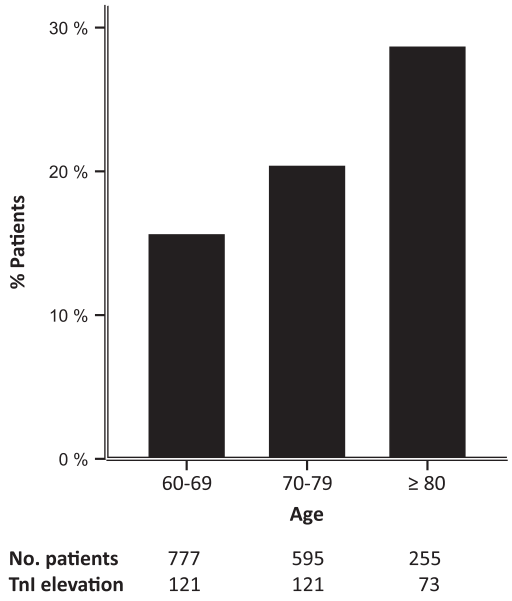


Figure 2. Distribution of myocardial injury over age. TnI indicates troponin I.

of preoperative factors known to be associated with postoperative death.[16] As suggested previously, implementation of postoperative monitoring of troponin as standard of care is feasible and may improve risk stratification and reclassification of patients at risk for early postoperative death, even when daily-care clinical protocols are not always followed.

This study is one of the first studies describing the results of implementation of routine postoperative monitoring of troponin as part of a standard postoperative care protocol. The study has some obvious limitations. First, because we measured troponin I only on the first 3 days after surgery, myocardial injury that may have occurred after that time interval was missed. However, previous research has shown that myocardial injury occurs primarily within the first 3 postoperative days.[18–20] Second, troponin I was not measured in all patients because it takes time to fully implement a new protocol in daily clinical practice. Patients in whom troponin I was not measured seemed somewhat healthier at baseline, and high-risk surgery was less frequent in these patients. In contrast, these patients without troponin I measurements more often underwent emergency surgery and reoperations with a higher incidence of death (Table 1). This apparent discrepancy of healthier patients who were more likely to die may be explained by the fact that troponin I measurements were more often (unintentionally) not ordered after emergency surgery. To determine whether these baseline differences resulted in a biased association between myocardial injury and death, we compared the results of the complete case analysis with the results of a sensitivity analysis including all patients after multiple imputations of the missing troponin I values. This analysis showed comparable results, indicating that imputation of missing data did not change the association between myocardial injury and short-term death. Third, because we did not measure troponin I before surgery, we could not adjust the results for preexisting troponin I elevations. Preexisting troponin I elevations may occur in patients with renal failure or other cardiovascular conditions. The cause of the higher preoperative levels of cardiac biomarkers in these patients is not completely clear but is thought to be caused by subclinical cardiac damage from silent myocardial ischemia or microinfarction, mild heart failure, left ventricular hypertrophy, or increased cardiac apoptosis.[21] We hypothesize that patients with a preexisting troponin I elevation may have an additional increase of troponin I because of ischemia caused by surgical stress. Although some patients with a preoperative acute coronary syndrome in whom troponin I levels decreased to normal values after surgery may have been missed, we do not think that this influenced the obtained results significantly. Moreover, although myocardial injury may already have occurred before surgery in some patients, postoperative troponin elevations may still predict short-term mortality.

The improved risk discrimination of patients with primarily asymptomatic myocardial injury, together with the time interval between troponin I elevation and death (median of almost 2 weeks), may potentially allow physicians to modify prognosis. First, we showed that in most patients with postoperative troponin I elevation, troponin I was only slightly (up to 10-fold) to moderately (up to 100-fold) increased, suggesting

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Table 2. Baseline Characteristics of Patients in Each Category of Troponin Elevation

	Troponin≤0.06 μg/L (n=1312)		Troponin 0.07–0.59 μg/L (n=258)		Troponin≥0.60 μg/L (n=57)		*P* Value
Male	660	(50.3)	148	(57.4)	36	(63.2)	0.03
Age, mean (SD), y	70.6	7.5	72.7	8.0	73.1	8.8	<0.01
RCRI factors							
High-risk surgery	372	(28.4)	103	(39.9)	26	(45.6)	<0.01
History of ischemic heart disease	192	(14.6)	53	(20.5)	19	(33.3)	<0.01
History of heart failure	21	(1.6)	11	(4.3)	3	(5.3)	<0.01
History of cerebrovascular disease	182	(13.9)	47	(18.2)	13	(22.8)	0.05
Renal failure	104	(7.9)	39	(15.1)	23	(40.4)	<0.01
Preoperative insulin use	66	(5.0)	18	(7.0)	8	(14.0)	0.01
Smoking*	226	(17.8)	50	(20.4)	10	(19.2)	0.61
Hypertension	649	(49.5)	148	(57.4)	32	(56.1)	0.05
Diabetes mellitus	221	(16.8)	51	(19.8)	16	(28.1)	0.06
COPD	134	(10.2)	28	(10.9)	5	(8.8)	0.89
Peripheral vascular disease	133	(10.1)	35	(13.6)	12	(21.1)	0.01
ASA class†							
1	174	(13.3)	21	(8.1)	2	(3.5)	<0.01
2	858	(65.4)	158	(61.2)	27	(47.4)	0.01
3	270	(20.6)	73	(28.3)	23	(40.4)	<0.01
4	10	(0.8)	6	(2.3)	5	(8.8)	<0.01
General anesthesia	1211	(92.3)	246	(95.3)	57	(100.0)	0.02
Emergency surgery	215	(16.4)	69	(26.7)	22	(38.6)	<0.01
Reoperation within 30 d	112	(8.5)	39	(15.1)	12	(21.1)	<0.01
Surgical specialty							
General	276	(21.0)	79	(30.6)	14	(24.6)	<0.01
Neurological	302	(23.0)	44	(17.1)	9	(15.8)	0.06
Vascular	186	(14.2)	47	(18.2)	22	(38.6)	<0.01
ENT and dental	169	(12.9)	33	(12.8)	4	(7.0)	0.43
Orthopedic	154	(11.7)	33	(12.8)	6	(10.5)	0.85
Gynecological	116	(8.8)	10	(3.9)	2	(3.5)	0.01
Urological	73	(5.6)	11	(4.3)	0	(0)	0.14
Plastic	36	(2.7)	1	(0.4)	0	(0)	0.03

Values are number of patients (%), unless indicated otherwise.
ASA indicates American Society of Anesthesiologists; COPD, chronic obstructive pulmonary disease; ENT, ear, nose, and throat; and RCRI, Revised Cardiac Risk Index.
*n=1271, n=245, and n=52, respectively, because of some missing values for smoking status.
†Physical status classification by the American Society of Anesthesiologists.

minor myocardial injury. In addition, only 3.2% of the patients with troponin I elevation had typical chest pain, and (minor) ECG changes suggestive of myocardial ischemia were found in only 9.5%. Still, these patients with minor myocardial injury had a higher risk of death. This risk seems comparable to the detrimental risk on survival of spontaneous acute coronary syndrome, suggesting a comparable pathophysiological mechanism in patients with isolated troponin elevation after surgery. Given the numerous plaque destabilizing and thrombogenic factors that are present in the perioperative period, troponin elevation may be a consequence of coronary plaque rupture with distal embolization of thrombus or, alternatively, thrombosis of small coronary arteries.[22–27] In these patients, further myocardial damage may be prevented by adequate treatment during the time interval between troponin elevation and death. Although many studies have addressed prevention of postoperative myocardial injury and death, a major shortcoming is the insufficiently resolved pathophysiology of such myocardial injury and POMI.[2–6] Myocardial injury may be caused by regional ischemia attributable to preexisting significant coronary artery disease or plaque rupture with thrombosis (acute coronary syndrome, type 1 myocardial infarction) or may be a consequence of diffuse myocardial ischemia attributable to an imbalance between myocardial oxygen supply and demand, caused by preexisting stable coronary stenosis or noncoronary factors such as deep anemia or hemodynamic instability (type 2 myocardial infarction).[1] The treatment benefits and possible harms in patients with postoperative myocardial injury remain to be evaluated, particularly those harms associated with invasive treatment options. A recent study

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Table 3. Association of Postoperative Troponin and 30-Day Mortality Adjusted for Age, Sex, Emergency Surgery, and Preoperative Renal Failure

	Unadjusted Analysis			Adjusted Analysis		
	RR	(95% CI)	*P* Value	RR	(95% CI)	*P* Value
Age (per year increase)	1.04	(1.01–1.07)	<0.01	1.01	(0.98–1.04)	0.33
Female sex	1.08	(0.64–1.80)	0.64			
Emergency surgery	5.76	(3.44–9.63)	<0.01	4.46	(2.62–7.58)	<0.01
Renal failure (preoperative)	2.67	(1.46–4.85)	<0.01	1.37	(0.74–2.53)	0.32
Troponin elevation, categorical						
Troponin ≤0.06 μg/L	Reference			Reference		
Troponin 0.07–0.59 μg/L	2.98	(1.66–5.34)	<0.01	2.36	(1.31–4.22)	<0.01
Troponin ≥0.60 μg/L	7.94	(4.07–15.5)	<0.01	4.19	(2.06–8.55)	<0.01

CI indicates confidence interval; and RR, relative risk.

in which 70 patients with postoperative troponin elevation were randomized to cardiology care consisting of monitoring, investigations, and changes to medications versus standard ward treatment showed no difference in 1-year mortality.[28] However, this study was performed in a limited number of patients.

The results of our evaluation of the data from routine postoperative monitoring of troponin are comparable to those obtained under research circumstances. Until recently, research was done in smaller cohorts of patients undergoing mainly vascular and other high-risk but elective surgical procedures, showing an incidence of postoperative myocardial injury of 8% to 52%.[4,12,13,29–39] Studies with a follow-up time of 12 months found a 2- to 20-fold increased mortality risk. Studies including emergency surgery reported an incidence of 33% to 53% of postoperative myocardial injury and 4- to 12-fold increased mortality risk.[40,41] We found a similar association in our routine troponin measurement cohort including elective and emergency surgery patients. As in previous studies, in most patients myocardial injury was found on the first postoperative day.[18–20] The recently published VISION study including >15 000 patients reported that elevation of troponin T after noncardiac surgery was strongly associated with 30-day mortality, with adjusted hazard ratios comparable to the adjusted relative risks of our protocol. The eligibility criteria of this study were comparable to those of our clinical protocol, except for the age criterion of >45 years instead of ≥60 years in our protocol. Thus, the predictive value of our standard of care postoperative troponin monitoring is markedly concordant with that found in a large cohort of patients included in the VISION study. The VISION investigators demonstrated that clinically relevant troponin thresholds existed below the value that corresponded to a 10% coefficient of variation of the troponin assay used. We considered the sample size of our study insufficient to confirm this finding. Nevertheless, the strengths of our study are that it focused on patients at higher ages and that patients from a single center were included within a shorter time period. In addition, we made an attempt to validate that the missing data were missing at random by imputation of missing troponin values. Moreover, in our study we used troponin I compared with troponin T in the VISION study. Because troponin I compared with troponin T is less influenced by renal dysfunction,[42,43] the results from our study may be more representative in patients with renal dysfunction. Finally, the VISION study did not include renal failure in the multivariable analysis, whereas in our study the association between postoperative myocardial injury and death was adjusted for preexisting renal failure.

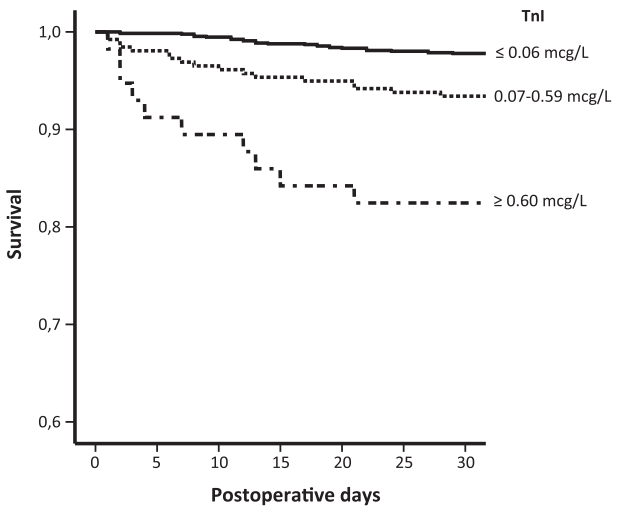


Figure 3. Survival curve of each category of troponin I (TnI) elevation.

We found a strong association between postoperative myocardial injury, as measured by troponin I, and death in noncardiac surgery patients. We therefore conclude that patients at risk for early death after noncardiac surgery are easily identified by routine postoperative troponin measurements. The time interval between troponin elevation and death potentially allows physicians to modify prognosis. For this purpose, the underlying pathophysiological mechanisms of postoperative myocardial injury must be explored. In addition, (1) whether there is a causal relationship between myocardial injury and mortality or whether myocardial injury merely indicates a worse outcome and (2) whether patients with myocardial injury may benefit from postoperative cardiovascular treatment must be elucidated.

Appendix

The Cardiac Health After Surgery (CHASE) Investigators include the following, associated with University Medical Center Utrecht, Utrecht, Netherlands: *Department of Anesthesiology:* Wolfgang F. Buhre, Professor of Anesthesiology; Jurgen C. de Graaff, Anesthesiologist; Cor J. Kalkman, Professor of Anesthesiology; Wilton A. van Klei, Anesthesiologist; Judith A.R. van Waes, Resident in Anesthesiology;

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Table 4. Association of Postoperative Troponin and 30-Day Mortality After Multiple Imputation of Missing Troponin Values and Adjusted for Age, Sex, Emergency Surgery, and Preoperative Renal Failure

	Unadjusted Analysis			Adjusted Analysis		
	RR	(95% CI)	*P* Value	RR	(95% CI)	*P* Value
Age (per year increase)	1.05	(1.02–1.07)	<0.01	1.02	(1.00–1.05)	0.12
Female sex	1.02	(0.65–1.58)	0.94			
Emergency surgery	6.87	(4.34–10.9)	<0.01	5.06	(3.09–8.25)	<0.01
Renal failure (preoperative)	2.63	(1.58–4.38)	<0.01	1.33	(0.77–2.29)	0.30
Troponin elevation, categorical						
Troponin ≤0.06 μg/L	Reference			Reference		
Troponin 0.07–0.59 μg/L	2.98	(1.66–5.34)	<0.01	1.93	(1.06–3.54)	0.03
Troponin ≥0.60 μg/L	7.94	(4.07–15.5)	<0.01	3.47	(1.67–7.21)	<0.01

CI indicates confidence interval; and RR, relative risk.

Leo van Wolfswinkel, Anesthesiologist. *Department of Cardiology:* Pieter A. Doevendans, Professor of Cardiology; Hendrik M. Nathoe, Cardiologist; Remco G. Grobben, Resident in Cardiology. *Department of Epidemiology:* Diederik E. Grobbee, Professor of Epidemiology; Linda M. Peelen, Epidemiologist. *Department of Clinical Chemistry and Hematology:* Hans Kemperman, Clinical Chemist; Wouter W. van Solinge, Professor of Clinical Chemistry. *Department of Radiology:* Tim Leiner, Radiologist. *Department of Surgery:* Gert Jan de Borst, Vascular Surgeon; Loek P.H. Leenen, Professor of Trauma Surgery; Frans L. Moll, Professor of Vascular Surgery.

Sources of Funding

This study was supported by a grant from the International Anesthesia Research Society (Clinical Scholar Research Award 2011 to Dr Van Klei), by a grant from the Friends of the University Medical Center Utrecht Foundation/Dirkzwager-Assink Fund to Dr Van Klei, and by departmental sources.

Disclosures

None.

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CLINICAL PERSPECTIVE

Perioperative adverse cardiovascular events are the leading cause of morbidity and mortality after noncardiac surgery. Despite efforts to prevent the occurrence of such events, the incidence of postoperative myocardial infarction is still high. To identify patients at risk for postoperative myocardial infarction, measuring cardiac troponin routinely after noncardiac surgery has been suggested. Therefore, in our hospital we implemented routine troponin measurement monitoring on the first 3 days after surgery as part of our standard postoperative care in patients aged ≥60 years undergoing intermediate- to high-risk surgery. This observational cohort study estimated the incidence of myocardial injury after noncardiac surgery and its predictive value on the risk of death within 30 days with the use of data from routine postoperative monitoring of troponin. Postoperative myocardial injury as detected by troponin elevation was found in 19% of the patients, and overall all-cause death within 30 days occurred in 3% of the patients. Although only a minority (<10%) of the patients had typical signs of myocardial ischemia or ECG changes, we found a strong association between postoperative myocardial injury and death. We therefore conclude that patients at risk for early death after noncardiac surgery are easily identified by routine postoperative troponin measurements. Furthermore, the time interval between troponin elevation and death potentially allows physicians to modify prognosis. Implementation of postoperative monitoring of troponin as standard of care is feasible and may improve risk stratification and reclassification of patients at risk for early postoperative death.

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on behalf of the Cardiac Health After Surgery (CHASE) Investigators

Circulation. 2013;127:2264-2271; originally published online May 10, 2013;
doi: 10.1161/CIRCULATIONAHA.113.002128
Circulation is published by the American Heart Association, 7272 Greenville Avenue, Dallas, TX 75231
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Print ISSN: 0009-7322. Online ISSN: 1524-4539

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